

# Rating Plausibility of Word Senses in Ambiguous Sentences through Narrative Understanding

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## Abstract

This report presents our final system for the AmbiStory word-sense plausibility prediction task from SemEval 2026 Shared Task 5. Unlike traditional Word Sense Disambiguation (WSD), which forces a single discrete “correct” meaning, this task models semantic interpretation as a graded phenomenon matching human consensus. We formulate the problem as a supervised regression task and fine-tune a RoBERTa-base encoder to predict human-perceived plausibility scores on a continuous scale from 1 to 5. To ground semantic interpretations, our system incorporates both dictionary definitions and usage examples for candidate senses. We evaluate our approach against a zero-shot Qwen2.5-3B-Instruct baseline using Spearman rank correlation ( $r_s$ ), Accuracy Within Standard Deviation ( $\text{Acc}_\sigma$ ), and Mean Absolute Error (MAE). Our fine-tuned RoBERTa model achieves an exact Spearman correlation of 0.4246 and an  $\text{Acc}_\sigma$  of 74.32%, substantially outperforming the zero-shot baseline across all metrics. The results demonstrate that task-specific fine-tuning is essential for modeling nuanced human plausibility judgments in ambiguous narrative contexts.

## 1 Introduction

Word Sense Disambiguation (WSD) is a fundamental Natural Language Processing task traditionally treated as a discrete classification problem, where a single meaning is assumed to be correct for each ambiguous word occurrence. However, natural language interpretation in real-world contexts is often inherently uncertain. Depending on contextual cues, background knowledge, and subjective viewpoint, multiple meanings may simultaneously appear plausible to human readers.

The AmbiStory shared task addresses this limitation by reframing WSD as a graded plausibility prediction problem. Rather than selecting a single sense, systems are required to predict how plausible a candidate interpretation appears to human

annotators on a continuous scale from 1 to 5.

Our objective was to investigate whether transformer-based architectures can approximate these nuanced human judgments. We hypothesized that although Large Language Models possess extensive linguistic knowledge, task-specific supervised learning is necessary to calibrate predictions to the particular rating scale, label variance, and narrative structure of the AmbiStory dataset.

## 2 Dataset Analysis and Problem Formulation

The AmbiStory dataset contains short narrative stories centered around ambiguous homonyms. The training split consists of 2,280 samples spanning 220 unique homonyms across 380 distinct story setups, while the development set contains 588 samples. Each sample provides:

- A three-sentence grounding precontext.
- An ambiguous sentence containing the target homonym.
- An optional story ending (No Ending, Ending A, or Ending B).
- A candidate sense represented as a dictionary definition (*judged\_meaning*).
- An example sentence illustrating that sense in isolation.
- Continuous human plausibility ratings averaged from at least 5 Prolific participants along with the standard deviation ( $\sigma$ ) of annotator ratings.

### 2.1 Label Distribution and Narrative Signal

Human plausibility ratings span the entire 1–5 range, with a mean score of 3.14 and a slight rightward skew. This distribution suggests that annotators often assign partial plausibility to candidate

senses rather than completely rejecting them. Consequently, a regression formulation is more appropriate than traditional binary classification.

We further quantified the influence of narrative endings by measuring the absolute divergence between competing sense scores ( $|\text{Sense A} - \text{Sense B}|$ ) across all 1,140 narrative conditions. The average divergence increases from 1.70 in no-ending conditions to 1.89 when endings are present, confirming that endings function primarily as semantic disambiguators rather than general plausibility amplifiers. They push one sense upward and the competing sense symmetrically downward.

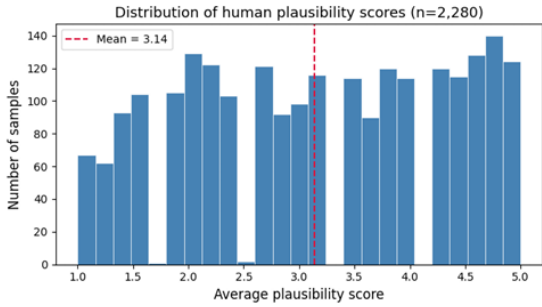


Figure 1: Distribution of human plausibility scores ( $n = 2,280$ ).

## 2.2 Human Disagreement

Analysis of annotator standard deviation reveals a pronounced inverted-V pattern: extreme plausibility scores (near 1 or 5) exhibit low disagreement ( $\sigma < 0.5$  for 462 samples), whereas mid-range examples (2.5–3.5) show substantially higher disagreement ( $\sigma > 1.5$  for 432 samples). This observation indicates that many samples are genuinely ambiguous rather than poorly annotated. Therefore, evaluation should account not only for absolute prediction error but also for inherent human uncertainty.

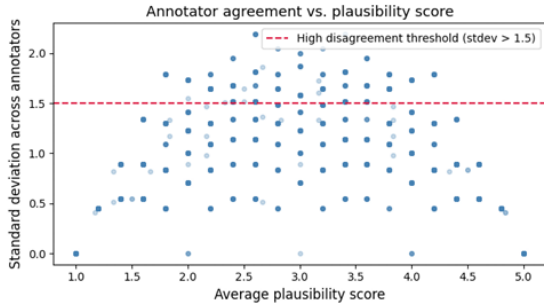


Figure 2: Annotator agreement vs. plausibility score.

## 2.3 Evaluation Metrics

Performance is evaluated using three complementary mathematical metrics:

**Spearman Rank Correlation ( $r_s$ ):** Measures the monotonic alignment between predicted and gold plausibility rankings:

$$r_s = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)} \quad (1)$$

**Accuracy Within Standard Deviation ( $\text{Acc}_\sigma$ ):** Evaluates whether predictions fall within the acceptable range defined by human disagreement:

$$\text{Acc}_\sigma = \frac{1}{N} \sum_i \mathbb{I}(|y_i - \hat{y}_i| \leq \max(\sigma_i, 1.0)) \quad (2)$$

where  $y_i$  denotes the gold score,  $\hat{y}_i$  denotes the predicted score, and  $\sigma_i$  represents the annotator standard deviation. The metric enforces a minimum tolerance window of 1.0 standard deviation.

**Mean Absolute Error (MAE):** Measures the average unweighted linear distance on the 1–5 rating scale:

$$\text{MAE} = \frac{1}{N} \sum_i |y_i - \hat{y}_i| \quad (3)$$

## 3 Technical Solution

### 3.1 RoBERTa Regression Architecture

Our primary system is based on RoBERTa-base, a pretrained transformer encoder containing approximately 125 million parameters. We replace the native masked language modeling layer and build a custom downstream regression head directly over the pooling position. The contextualized final hidden representation produced by the [CLS] token is passed through a two-layer Multi-Layer Perceptron (MLP) incorporating intermediate Gaussian Error Linear Unit (GELU) activations, an explicitly tuned dropout regularization layer (0.2), and a final linear projection layer to output a single continuous scalar score.

### 3.2 Input Representation

Accurate plausibility estimation requires integrating information from multiple textual components. We serialize each sample into a single tokenized sequence using delimiter tokens:

```
[CLS] Precontext [SEP] Ambiguous
Sentence [SEP] Ending [SEP]
Judged Meaning: Example Sentence
[SEP]
```

When an ending condition is absent, the corresponding segment remains completely empty. This structure allows RoBERTa’s multi-head self-attention mechanism to build cross-attentional representations between the homonym’s surface form and its dictionary definition, grounding candidate senses with explicit semantic information.

### 3.3 Training Objective and Optimization

The model is optimized using Mean Squared Error loss:

$$\text{Loss}_{\text{MSE}} = \frac{1}{N} \sum_i (y_i - \hat{y}_i)^2 \quad (4)$$

The final linear scalar is mapped to the target rating range via a scaled sigmoid function:  $\hat{y} = 1.0 + 4.0 \cdot \sigma(x)$ . During inference, continuous float predictions are converted into integer outputs for final formatting using nearest-neighbor rounding and clipping strictly to the  $[1, 5]$  interval.

Training was performed using AdamW optimization with a linear learning-rate schedule. Hyperparameters were selected via an extensive empirical grid search over an internal validation partition:

| Parameter     | Value                 |
|---------------|-----------------------|
| Learning Rate | $2 \times 10^{-5}$    |
| Batch Size    | 16                    |
| Epochs        | 10                    |
| Warmup Ratio  | 10% of steps (linear) |
| Weight Decay  | 0.01                  |
| Dropout Rate  | 0.2                   |

Table 1: Hyperparameters for RoBERTa fine-tuning.

## 4 Evaluation Results

### 4.1 Quantitative Performance

Table 2 compares the fine-tuned RoBERTa model against the zero-shot Qwen2.5-3B-Instruct baseline evaluated on the 588-sample development set. For the zero-shot pipeline, predictions were extracted as a logit-weighted continuous mathematical expectation over the numeric tokens ‘1’ through ‘5’ to capture calibrated model uncertainty.

The fine-tuned RoBERTa model consistently and substantially outperforms the zero-shot baseline across all metrics. The 87% relative gain in Spearman correlation confirms that task-specific supervision drastically improves the model’s capacity to mirror human plausibility orderings.

## 5 Comprehensive Error Analysis

### 5.1 Disagreement Tier Analysis and Calibration Diagnostics

To uncover underlying model behaviors, we stratified the 588 validation samples into three equal-sized tiers by annotator standard deviation (Low, Mid, and High Disagreement). Our analysis revealed a critical, counter-intuitive pattern in the zero-shot pipeline:

- Low Disagreement Tier:  $\text{MAE} = 1.4888$ ,  $\text{Acc}_\sigma = 5.88\%$
- High Disagreement Tier:  $\text{MAE} = 0.6946$ ,  $\text{Acc}_\sigma = 92.63\%$

This disparity exposes an evaluation artifact of the  $\text{Acc}_\sigma$  metric. When human disagreement is high, the standard deviation expands the tolerance window so wide that virtually any prediction falls inside it, making the metric trivially easy to satisfy. On low-disagreement samples where annotators converged on a narrow consensus range, the zero-shot model completely failed to calibrate to the targets.

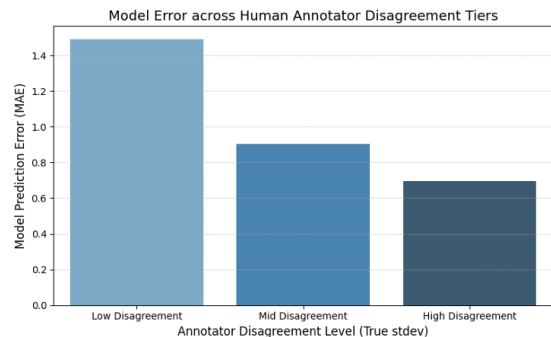


Figure 3: Model error across human annotator disagreement tiers (Zero-Shot).

The fine-tuned RoBERTa model successfully corrected this calibration failure, achieving stable and balanced error boundaries across all ambiguity levels:

- Low Disagreement Tier:  $\text{MAE} = 1.05$
- Mid Disagreement Tier:  $\text{MAE} = 0.82$
- High Disagreement Tier:  $\text{MAE} = 0.85$

Supervised fine-tuning allows the network to maintain its predictive precision in clear consensus cases while preserving appropriate caution when handling highly subjective or ambiguous text.

| Metric              | Qwen2.5-3B (Zero-Shot) | RoBERTa (Fine-Tuned) | Relative Gain | Interpretation                   |
|---------------------|------------------------|----------------------|---------------|----------------------------------|
| Spearman ( $r_s$ )  | 0.2264                 | 0.4246               | +87%          | Moderate positive rank alignment |
| Acc $_{\sigma}$ (%) | 51.53%                 | 74.32%               | +44%          | Strong scale calibration         |
| MAE                 | 1.0554                 | 0.9163               | +13%          | Lower average residual error     |

Table 2: Performance comparison — zero-shot baseline vs. fine-tuned RoBERTa.

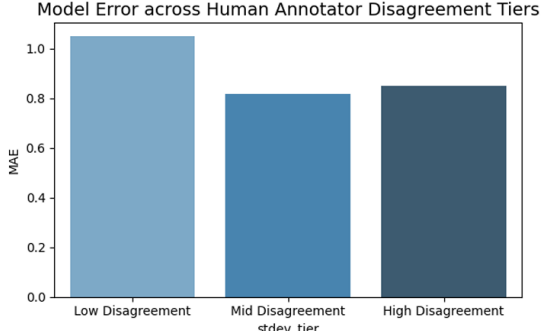


Figure 4: RoBERTa MAE across human annotator disagreement tiers.

## 5.2 Confusion Matrix and Systematic Variance

A systematic look at RoBERTa’s prediction distributions highlights specific areas of difficulty:

**Prediction Compression (Conservatism):** The fine-tuned model exhibits a visible regression-to-the-mean effect. When the true human rounded score is a maximum of 5, RoBERTa predicts 5 only 17 times, opting for a conservative rating of 4 in 48 instances. Similarly, it predicts an extreme low score of 1 only 14 times across all 588 samples.

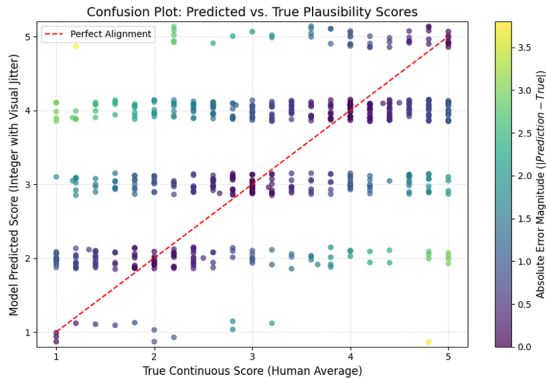


Figure 5: Confusion plot — predicted vs. true plausibility scores.

**Per-Homonym Vulnerability:** Errors are heavily concentrated on specific abstract or highly polysemous vocabulary items. Words like “blaze” (MAE  $\approx$  2.0), “try” (MAE  $\approx$  1.9), and “croaked”

(MAE  $\approx$  1.8) produce the highest errors. These terms represent instances where context remains fundamentally incomplete, causing predictions to diverge.

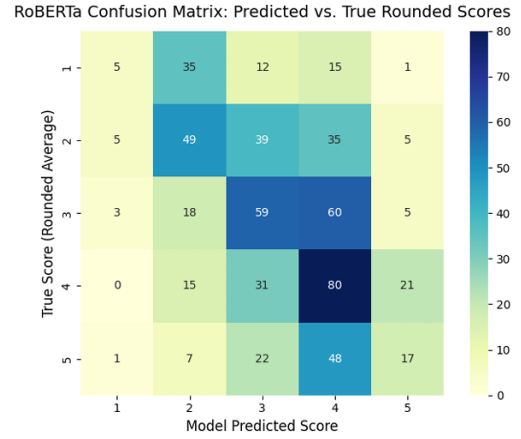


Figure 6: RoBERTa confusion matrix — predicted vs. true rounded scores.

## 5.3 Qualitative Cases

**Successful Example:** For the homonym “dribbling,” football-related contextual tokens allowed the attention heads to build a tight cross-attentional map with the sporting dictionary entry, predicting a high plausibility score of 4.5, matching the human judgment of 4.6.

**Failure Example:** For the homonym “track” in a no-ending narrative context, the model overestimated the plausibility of one sense (predicting 4.1) where human raters remained highly uncertain and split (averaging 3.0). This highlights a tendency toward overconfidence when handling ungrounded ambiguous contexts.

## 6 Reflection

Our final project implementation deviated from our original Milestone 2 proposal in three distinct ways:

**Validation Design Strategy:** We initially designed an internal validation partition using a GroupShuffleSplit by homonym to prevent the

model from utilizing homonym-specific shortcut strategies. However, data analysis later confirmed that the official SemEval train and development splits were already completely disjoint, with zero overlapping homonyms. This allowed us to leverage the entire training set for model optimization while maintaining absolute architectural generalization testing on the provided development set.

**Feature Engineering over Model Scaling:** Our initial plan prioritized comparing a large suite of diverse encoder architectures (BERT, RoBERTa, DeBERTa). Experiments quickly revealed that optimizing input representation layouts—specifically adding dictionary glosses and concrete usage examples—offered much larger performance gains than changing the underlying base model encoder.

**Output Post-Processing Integration:** While our network natively runs continuous regression to model human averages, we integrated a rounding and clipping post-processing pipeline during inference to meet the official integer output specifications without losing our regression loss advantages during training.

## 7 Limitations and Future Work

Several limitations remain in the current framework. First, the 588 development samples represent a relatively small set for evaluating fine-grained variations. Second, the continuous target labels represent aggregate human averages that collapse multi-modal distributions; the model does not explicitly output prediction uncertainty or model annotator disagreement variance.

Future work should investigate deep ranking-consistent ordinal regression objectives (such as CORN) to directly optimize for Spearman rankings rather than unweighted linear distances. Additionally, exploring uncertainty-aware architectures or ensembling larger open-weight models (e.g., Qwen2.5-14B fine-tuned via LoRA) could address remaining calibration issues in highly ambiguous contexts.

## 8 Conclusion

This project evaluated graded word-sense plausibility prediction on the AmbiStory task using a supervised RoBERTa-base regression framework. Our experiments show that task-specific fine-tuning significantly outperforms zero-shot prompting with

larger language models on continuous human rating targets. Furthermore, our results highlight the importance of contextual grounding by explicitly providing dictionary definitions and example sentences. While modeling subjective human judgments remains a challenging task, contextualized transformer representations provide an effective foundation for navigating semantic ambiguity.

## References

- Janosch Gehring and Michael Roth. 2026. Rating Plausibility of Word Senses in Ambiguous Sentences through Narrative Understanding. *Proceedings of the 20th International Workshop on Semantic Evaluation (SemEval-2026)*. Association for Computational Linguistics.
- Yinhan Liu, Myle Ott, Naman Goyal, Jingfei Du, Mandar Joshi, Danqi Chen, Omer Levy, Mike Lewis, Luke Zettlemoyer, and Veselin Stoyanov. 2019. RoBERTa: A Robustly Optimized BERT Pretraining Approach. *arXiv preprint arXiv:1907.11692*.
- Daniel Jurafsky and James H. Martin. *Speech and Language Processing*.
- Xintong Shi, Wenzhi Cao, and Sebastian Raschka. 2023. Deep neural networks for rank-consistent ordinal regression based on conditional probabilities. *Pattern Analysis and Applications*, 26:941–955.